

Cumin (*Cuminum cuminum* L.): Present breeding status and future prospective

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ABSTRACT

Cumin is an important spice crop with various medicinal properties. It is a potential source of bioactive compounds with antidiabetic and anticancer activity. It is predominately grown in the arid and semi-arid regions of the world. Cumin is one of the most widely used herbal drugs in the Indian (Ayurveda) system of medicine. Therefore, high-yielding, disease-resistant cumin genotypes with better-quality attributes are in demand today. The lack of advanced technologies for the production of desirable genotypes is one of the major constraints in the cumin improvement program, which can be achieved by adopting modern integrated breeding technologies. Although many conventional and biotechnological approaches are being used regularly to develop an improved variety of cumin, less effort has been made in the use of integrated breeding techniques. This paper deals with the comprehensive botanical aspects, genetic resources, and the biotechnological approach to improve traits of cumin. Review underline research attempts made in cumin in various aspects including collection, conservation of genetic resources, mutation breeding, selection breeding, tissue culture, molecular markers, genetic transformation, and gene editing. In this paper, an attempt has been made to analyze holistic strategy for cumin improvement and enhance the production. Future research preventatives in this direction are also suggested.

1. Introduction

Cumin (*Cuminum cyminum* L.) is one of the most important spice crops and medicinal plants in the world, which belongs to the family Apiaceae (Umbelliferae). It originated mainly in Egypt five million years ago and is widely grown under hot, dry tropical, and subtropical regions of the world, primarily in India, China, Iran, Syria, Turkey, United Arab Emirates and Egypt (Lodha and Mawar, 2014). India is a chief producer as well as an exporter of cumin to Brazil, Japan, UK, US UAE, etc., and plays an important role in terms of area, production, export, and consumption. It is cultivated on an area of 8.42 lakh hectares with a total production of 5.47 lakh tonnes in India, and has 70 % of the share in the global market (Dar et al., 2019). In Indian states, Rajasthan and Gujarat are the leading producers, contributing about 90 % of the total cumin production, and it is also cultivated in another state as a Rabi season crop. The total cumin export continuously increased (1.84 lakh metric tons in 2023–24) from 2010 to 2024. A major reason for the high demand for cumin in the global market is due to its extensive applications in herbal medicine and cooking. (Singh et al., 2017; Meena et al., 2021). Seeds of cumin are known for their medicinal properties and aroma;

therefore, they are used in many Ayurvedic and veterinary medicines (Kumar et al., 2021). It is found that antioxidant properties are mentioned about its astringent, carminative, and coagulant behavior for remedies against various diseases like diarrhea, dyspepsia, epilepsy, toothache, whooping cough, flatulence, indigestion, and jaundice (Singh et al., 2017; Piri et al., 2019).

However, potential medicinal use of cumin for human benefits has created a huge demand in the global market. but, the availability of high-yielding varieties for its medicinal use is still limited. The limitation of diverse alleles is the main constraint to the quality production of spice crops. Moreover, the growth and development of crops are also affected by various stresses in their life cycle (Adil and Quraishi, 2023). Numerous conventional breeding methods are routinely used in the improvement of cumin (Johnny and Ravindran, 2006; Piri et al., 2019). While the major constraints of cumin crop cultivation are low genetic diversity with a narrow genetic base, which cannot be efficiently overcome by conventional breeding methods (Bahraminejad et al., 2011). Therefore, cultivation practices of cumin require improved agronomic tools and proper management to get a high-quality yield. However, the shortage of superior varieties with desirable agronomic as well as

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Table 1;

The Modified table consists of the number of cumin genotypes conserved and characterized in different organizations (Meena et al., 2011; Tyagi et al., 2004).

Organization	Accessions (no.)
NBPGR, New Delhi	8
Rajasthan Agricultural University, Research Station, Jobner (Rajasthan)	218
Gujarat Agricultural University, Jagudan (Gujrat)	59
NOUAT, Kumarganj (Ayodhya)	19
National research center on seed spices, Ajmer (Rajasthan)	143
NBPGR research station Jodhpur (Rajasthan)	55
Leibniz Institute of Plant Genetics and crop Plant Research (Germany)	17

Table 2

List of genotypes improved via different breeding methods (Johny and Ravin-dran, 2006).

Culture	Pedigree/method	Properties
Mc-43	selection of germplasm	moderately tolerant to Fusarium wilt, Alternaria blight and powdery mildew, Average yield: 5.8 q ha ⁻¹
GC-I	selection from vijaypuram-5	Wilt tolerance, Average yield: 7.0 q ha ⁻¹
RZ-19	Recurrent selection in UC-19	tolerant to wilt and blight, Average yield: 5-6 q ha ⁻¹
GC-2	Induced mutation in NIC-49	Average yield: 7.0 q ha ⁻¹
GC-3	Recurrent selection from German germplasm EC-232689	Wilt resistant, essential oil content 3.5 %, Average yield: 7.0 q ha ⁻¹
S-404	selection of germplasm	moderately resistant to powdery mildew disease, Average yield: 3.5 q ha ⁻¹
GC-4	selection of germplasm	Resistant to Fusarium wilt, Average yield: 8.75 q ha ⁻¹
RZ-209	Recurrent selection from Jalore	Wilt tolerance and Average yield: 6.5 q ha ⁻¹
RZ-223	Mutation selection from UC-216	Wilt tolerance, essential oil content 3.2 % Average yield: 6.0 q ha ⁻¹
AC-01-167	Re-selection from EC-243375	resistance to Fusarium wilt, Average yield: 5.15 q ha ⁻¹
RZ-345	recurrent selection	high volatile oil content and lesser infestation of wilt, blight and powdery mildew, Average yield: 6.07

economic traits can be only fulfilled by the use of modern techniques (Kumar et al., 2023). In addition, integrated advanced breeding tools for cumin improvement are not discussed by earlier authors. The present review describes how the recent development of genome editing tools can be utilized in the future for the traits improvement program and commercial production of cumin. Biotechnological tools such as micropropagation, somaclonal variation, *in vitro* conservation, synced technology, protoplast fusion, and transgenic development have great potential for cumin improvement programs. Hence, there is an urgent need to compile the data on the utilization of biotechnological tools for cumin improvement. The present review collects information related to botany, morphology, utilization of modern tools, and improved varieties of cumin crops. In addition, many novel breeding tools have been discussed for cumin improvement, and major thrust areas in the research are also discussed in this paper.

2. Botany and growth of cumin

Cumin (*Cuminum cyminum* L.) is an annual, diploid, herbaceous plant. It consists of a slender, branched stem of 30–40 cm tall and 3–5 cm in diameter (Kumar et al., 2023). The stem is grey or dark green and consists of branches with 2–3 sub-branches that form the uniform

canopy. The leaves are pinnate or bipinnate, with thread-like leaflets 5–10 cm in length. The flowers are small in size, borne in umbels and pink or white in color. Each umbel consists of five to seven small umbellate. The fruit of cumin is an ovoid achene 4–5 mm long or lateral fusiform, containing a single seed with two pericarps. The seed of cumin is elongated, yellowish brown, and has eight ridges with oil canals (Chandola et al., 1970).

2.1. Habitat, distribution, and climate condition

Cumin crops are mainly grown in 20°N–38°N latitudes of tropical and subtropical regions. It requires a temperature range from 9°C to 30°C and low atmospheric humidity. However, high humidity is deleterious during flowering and seed setting as the crop is susceptible to many fungal diseases (Kafi, 2006). Cumin thrives generally in sandy soil with low organic matter. Seed sowing occurs between mid-November and the first week of December with a 100–120-day maturity period. Globally, India and Iran are the prime producers of cumin. It is also grown in Argentina, Egypt, Morocco, Ukraine, Malta, Mexico, Lebanon, Pakistan, Afghanistan, Turkey, Central America, and Central Asia (Kafi et al., 2006). In India, it is mainly grown in Rajasthan, Gujarat, Tamil Nadu, and, to some extent, in Uttar Pradesh, Punjab, and Madhya Pradesh.

2.2. Genetic resources and diversity

Many authors observed that cumin consists of low genetic divergence and genetic resources, which are due to more disease infestation, lower crop productivity (Meena et al., 2023), and limited availability of the potential gene pool. There is an urgent need to increase germplasm exploration, characterization, and evaluation methods for successfully established cumin crops under diverse climate conditions. Various exploration missions were undertaken by different research organizations, mainly in the Indian states of Rajasthan and Gujarat. The research institute successfully established *in-situ* and *ex-situ* methods for the conservation and multiplication of the genetic stock of cumin. Methods of conservation mainly included seed genebank, DNA storage, pollen storage, field genebanks, botanical gardens, etc. (Meena et al., 2011; Tyagi, 2004). The number of accessions being collected and maintained at various collection centers is mentioned in Table 1.

3. Crop improvement

The increasing global demands of spices have various challenges, mainly related to quality traits, production, and lack of genetically improved genotypes. India is one of the main producers of spices globally (<https://www.seair.co.in/blog/cumin-seed-export-from-india.asp>). As feed for the growing population, the demand for the production of seed spices increases day by day. Additionally, the introduction of stress-tolerant varieties of cumin is equally important for fulfilling the requirements of the growing population. The availability of literature gives narrow information about constraints in cumin improvement, like a limited genetic base in germplasm for yield, quality, and volatile oil content (Kumar et al., 2015). Conventional plant breeding, including wide crosses, the introduction of desirable traits from wild relatives by backcrossing, double haploid technology, heterosis, recurrent selection, mutation, polycross, etc., are breeding patterns that are involved in overcoming these constraints (Marthe, 2018). Plant breeders mainly focus on the improvement of cereals. As cumin is a spice crop, the study of its complex characteristics inheritance is very limited. Therefore, there is a need to emphasize the initiation of a hybridization program for the improvement of cumin genotypes.

Work based on heterosis breeding revealed that higher phenotypic distance in genotypes is due to allelic differences for polygenic traits (Tomkowiak et al., 2020). The essence of heterosis has been reported in fennel (Verma and Saxena, 2024) and coriander (Singh and Ramanujam, 1972). Improvement in spices is the process of introducing multiple

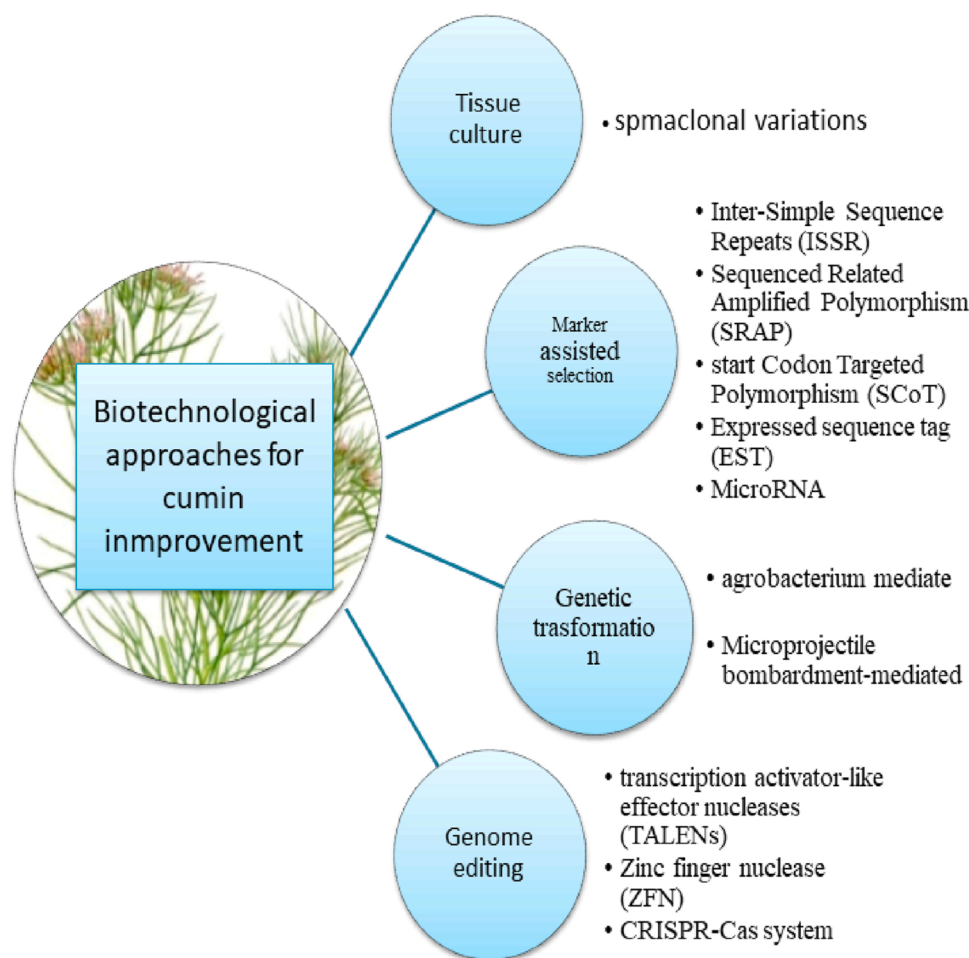


Fig. 1. Representative figure of various approaches used for cumin improvement program.

Table 3;

Major Phytochemicals of cumin essential oil according to International Standards Organization (ISO: 9301/2003) and their health benefits (Nouri et al., 2024; Mohammed et al., 2024).

Sr. No.	Bioactive compound	Percentage (%)	Properties
1	Cuminaldehyde	21.31–33.60 %	Exhibits antimicrobial, antioxidant, anti-inflammatory, antidiabetic, neuroprotective activities and elevated milk production
2	p-Cymene	3.16–7.89 %	Antioxidant, anti-inflammatory, antiparasitic, antidiabetic, antiviral, antitumor, antibacterial, and antifungal activities
3	γ-Terpinene	13.68–23.29 %	Antioxidant and antibacterial and immune-boosting activities
4	β-Pinene	9.32–11.46 %	Reduce inflammation and exhibits antimicrobial property
5	p-mentha-1,3-dien-7-al	10.06–14.02 %	Exhibits antimicrobial, antioxidant, anti-inflammatory effects
6	p-mentha-1,4-dien-7-al	14.44–20.84 %	Exhibits antimicrobial, antioxidant, anti-inflammatory effects

positive genes through selection or recurrent selection and pyramiding, which results in the development of desirable genotypes. This approach has produced several good varieties of cumin, fennel, coriander, ajwain, and sowa. (Hore, 1979). The appropriate strategy in Apiece

(cross-pollinated) spices have the objective of improving the specific populations through mass selection and recurrent selection based on the performance of individual plant progenies. Genetic enhancement of cumin can also be possible through the process of induced mutation. It provides a better option to develop natural variability in the cumin. Variation for different traits like flowering period, days to maturity, and individual weight of the cumin plant, etc., can be induced by gamma rays. Dry seeds of cumin (cv. RZ-19) were irradiated with 20, 30, 40, 50, and 60 kR gamma radiation using Cobalt-60 to induce variability (Sastri and Sharma, 2011; Koli et al., 2000).

Polycross methods are used to improve the performance of synthetic varieties, which are the result of crossbreeding between several selected parents. Different methods, such as general combining ability, heterosis, and inbreeding depression, are used to select the superior genotypes, and genetic diversity among the parents in polycross can significantly affect the success of new varieties (Reed and Frankham, 2003). Hence, optimization of genetic diversity is important in a plant breeding program. Different types of breeding methods involved in the improvement of cumin varieties are shown in Table 2.

3.1. Biotechnological approach

Like other spices, crop yield and yield-related components of cumin are also affected by various biotic and abiotic stresses (Kumar et al., 2021). Therefore, improvement in cumin genotypes is required to prevail over these kinds of stresses. Although different conventional breeding methods are routinely used for the selection of superior genotypes, they are not sufficient to overcome biotic and abiotic

Table 4

Merits and demerits of the various breeding techniques for cumin improvement.

Methods / techniques	Traits improved	Limitation	References
conventional breeding			
a. Genetic diversity and conservation	Diversity, conservation and evaluation of Cumin	Genetic erosion of germplasm and seed become inviable.	Meena et al. (2011); Tyagi et al. (2004)
b. Heterosis	Yield and yield related traits in Spices crops	only phenotypic traits are considered in crop improvement.	Verma and saxena (2024); Singh and Ramanujam (1972)
C. Recurrent selection	Population improvement for yield related traits of cumin, fennel, coriander, ajwain, and sowa.	selection breeding is time consuming	Hore (1979)
d. induced mutation	Induced variability by physical mutagen in Cumin	desired mutations are often rare and undesirable mutation can occur	Sastry and Sharma (2011); Koli et al. (2000)
Biotechnological approaches			
a. Tissue culture	Regeneration of cumin via callusing	Bacterial and other microorganism contamination. less resilient to diseases when grown in outside conditions	Mamdouh and Smetanska (2022); Woo et al. (2021)
b. Somaclonal variation	Induced variation in nuclear as well as cytoplasmic genomes of cumin	Unpredictable variation and genetic unitability.	Nwauzoma and Jaja (2013); Anil et al. (2018); Ahmed et al. (2011)
c. Marker Assistant Selection	Evaluate genetic fidelity of cumin plants	can lead to selecting the wrong plant and low reliability.	Martínez-Estrada et al. (2017); El-Fattah and Haridy (2019); El-Shahed et al. (2017); Bhatt et al. (2017)
d. MicroRNAs	Biosynthesis of piperine in black pepper, identification of biosynthesis pathway of terpenoid in spices crops	Complexity in stress-associated regulatory networks of MicroRNA.	Ding et al. (2021); Reyes et al. (2023)
e. Proteomics	Identification of α -amylase inhibitory, antioxidant and antidiabetic activities in Cumin	complexity of plant proteomes and it is challenging to interpreting large datasets	Pandey et al. (2016)
f. Genetic Transformation	Improvement of saline resistance genotypes	taking a longer time to transfer desirable traits and integrates DNA at any random sites in the genome.	Pandey et al. (2016)
g. Genome editing	Develop climate-resilient genotypes, induce expression of secondary metabolite.	expensive difficult to perform.	Angon et al. (2023); Aparna (2023)

constraints (Fig. 1). However, different conventional breeding methods are routinely used for the selection of superior genotypes. They are not sufficient to overcome biotic-abiotic constraints and cannot improve overall nutrition with multiple health benefits. Therefore, the intergradation of conventional as well as biotechnological tools is required to enhance the nutritional value and global production of cumin (Fig. 3).

3.1.1. Tissue culture (micropropagation/clonal selection)

To meet the global demand for crop spice, its large-scale cultivation has become a primary objective of plant breeders. Plant tissue culture techniques provide superior planting material for cumin for commercial cultivation (Aruna and Sivaramakrishnan, 1996). Additionally, cumin propagated by seeds is generally time consuming and labor-intensive. Therefore, the cell culture technique is now popularly and successfully used in the culture of various species of cumin (Hussien et al., 2022). However, the major problem is identifying pure traits of cumin grown under field conditions. This constraint may be overcome via direct regeneration of cumin (Ebrahimie et al., 2007) and through embryo culture (Tawfik and Noga, 2002; Valizadeh, 2018). Other authors studied different developmental stages of the embryo of cumin under various media conditions (Hussein and Batra, 1998). Shoot regeneration via somatic embryogenesis techniques is also used for large-scale production of cumin (Tawfik and Noga, 2001). Adventitious shoot proliferation from sterile germinated seedlings of *Cuminum cyminum* was also reported (Bahraminejad et al., 2011). This study supports the conservation of cumin germplasm by direct regeneration of plants from nodes and embryos. Cumin regeneration via callus induction is also an effective technique for the mass multiplication of cumin in a short period (Mamdouh and Smetanska, 2022; Woo et al., 2021), and plant regeneration from the callus of *C. cyminum* in cell suspension culture are also documented. Several studies have been conducted based on shoot regeneration and callus induction (Ebrahimie et al., 2007; Kiani and Siahchehreh, 2018).

Genetic variations in crop plants are one of the effective sources of providing diverse genotypes that are used in future breeding programs. Somaclonal variation is a common, important pathway that leads to changes under tissue culture conditions. (Larkin and Scowcroft, 1981;

Nwauzoma and Jaja, 2013). These processes mainly involve changes in nuclear as well as cytoplasmic genomes (Anil and Lobo, 2018; Ahmed et al., 2011). Somaclonal variation is described as genetic changes either at the chromosomal level, like changes in chromosome number and structure, or at the level of DNA sequence, like point mutations, or modification in methylation patterns of DNA and histones. Cytogenetic abnormality in a number of chromosomes in cumin genotypes was also found (Bharti et al., 2018). However, the lack of improved genotypes via somaclonal techniques is far-reaching and does not seek the attention of plant breeders. Therefore, in the future, new possibilities will be available to explore the improved genotypes of cumin through somaclonal variants.

3.1.2. Marker assistant selection, genomics and proteomics

Cumin has been used as a medicinal plant for a long time; however, the lack of their genetic variability becomes a limiting factor in the selection of suitable genotypes for a breeding program. It is widely known that the selection of suitable genotypes is mainly based on phenotypic traits rather than genotypic traits. Additionally, it is very difficult to manage several populations of cumin using traditional approaches due to the difficulty in measuring complex traits based on environmental factors. Molecular marker-based selection is a powerful tool for the assessment of genetic diversity besides from the environment (Martínez-Estrada et al., 2017). Molecular markers like Sequenced Related Amplified Polymorphism (SRAP) and Inter-Simple Sequence Repeats (ISSR) have been suggested to be convenient methods for assessing the genetic fidelity of cumin plants (El-Fattah and Haridy, 2019; El-Shahed et al., 2017; Pradeep et al., 2002; Bhatt et al., 2017). The low genetic diversity of cumin is easy to identify without prior knowledge of these DNA sequences (Mohamamadzadeh et al., 2022). Some of the complex traits of cumin genotypes were detected by molecular marker analysis (Hussien et al., 2022). Similarly, diversity among various genotypes of cumin was examined using ISSR markers, and it is an effective and reliable marker for the selection of a wide genetic base. Other markers, like the start Codon Targeted Polymorphism (SCoT) marker revealed polymorphism in cumin (Barboza et al., 2018). EST sequencing gives information about the putative

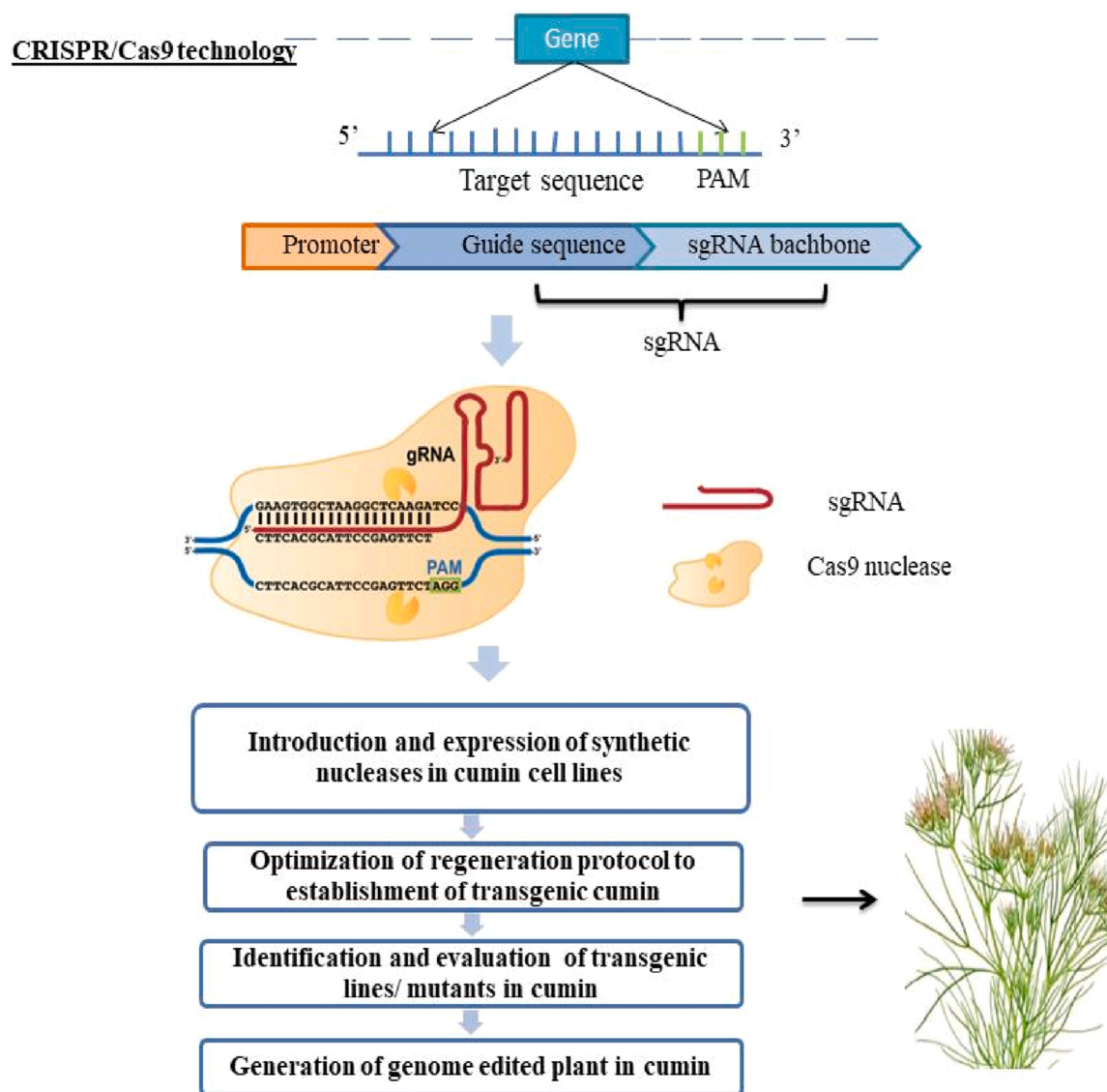


Fig. 2. Representative process diagram of cumin improvement using genome editing tools.

function of genes expressed in different developmental stages of species (Armenta-Medina and Gillmor, 2019). Currently, NCBI-EST databases (www.ncbi.nlm.nih.gov/dbEST/) are available for many species of spice crops like ginger, curcuma, black pepper, and garlic.

MicroRNAs (miRNAs) are short non-coding sequences of RNA (18–23 nucleotides) that occur in various species of plants, animals, and some viruses that control the expression of genes at the post-transcriptional level by the process of inhibition of mRNA. Plant miRNAs play an important role in plant growth and survival of plants against abiotic and biotic stress responses. In a study, it was found that many specific microRNAs are involved in plant secondary metabolism production. Hence, in future studies, it could support the production of plant secondary metabolites by miRNA-mediated transgenic research regarding commercial-scale cultivation of cumin. (Reyes-Calderón et al., 2023) Medicinally important crops mainly synthesize secondary metabolites during different developmental stages. Recently, tmiRNA-based transformation technology has been used for the optimization of genes responsible for the biosynthesis pathway of secondary metabolites in many spice crops. In the studies of *C. cyminum*, the biosynthesis of secondary metabolites associated with the miRNA profile was determined using NGS technology (Chand et al., 2016). It is observed that the incidence of soft rot disease in turmeric occurs due to

differential expression of these miRNAs and their targets (Ding et al., 2021). MicroRNA is involved in the biosynthesis of piperine in black pepper (Chen et al., 2018). In a study using a high-throughput Illumina-based sequencing platform, 353 conserved and 39 novel cumin miRNAs are identified. Among the identified miRNAs, one novel miRNA was involved in the biosynthesis pathway of the carotenoid, while 14 conserved miRNAs were found to be specific enzymes that were significantly expressed in the biosynthesis pathway of terpenoid (Chand et al., 2016).

Major protein fractions and cumin protein isolates (CPI) were extracted from cumin seeds. Functional properties, structure, and physicochemical properties of proteins were observed (Siow and Gan, 2017). The protein content of cumin seeds plays an important role in potential α -amylase inhibitory, antioxidant activities, and antidiabetic activities. Albumin, globulin, and glutelin are the predominant protein fractions of cumin seeds, identified to be responsible for their bio-activities (Umadevi et al., 2018). Plant antimicrobial peptides (AMPs) are essential components in the defense system of plants. AMPs are expressed under certain stress conditions. Involvement of abundant AMP in black pepper and their expression against *Phytophthora* infection is also described using a label-free quantitative proteomics strategy (Singh et al., 2010). 24 AMPs belonging to the cationic, anionic,

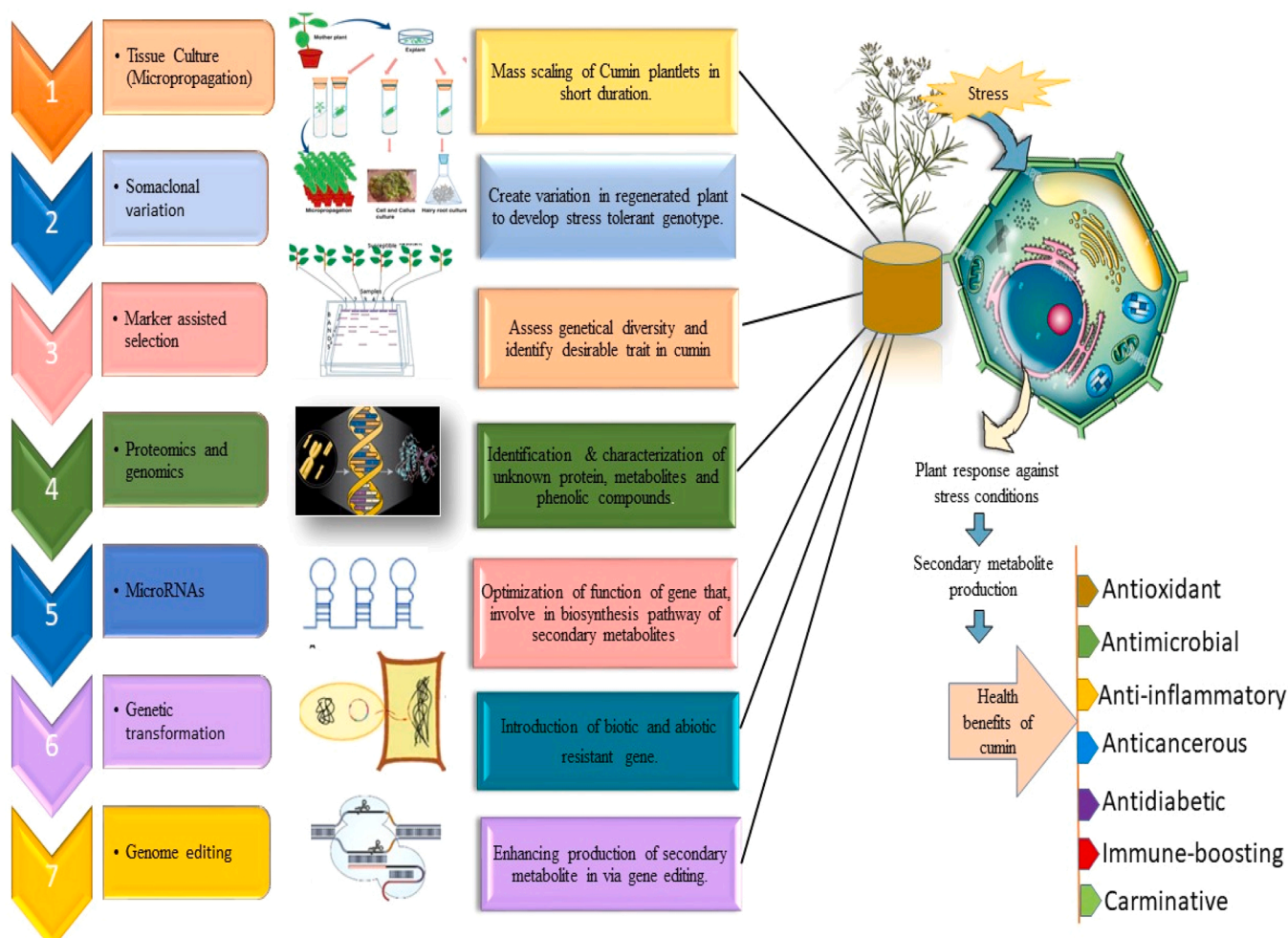


Fig. 3. Various breeding tools and its effect on metabolic pathway for quality enhancement of cumin.

cysteine-free, and cysteine-rich groups are working in the defense network of black pepper. This report gives an idea about the use of AMPs against pathogens in other species of cumin.

3.1.3. Genetic transformation

Genetic transformation is a procedure for the introduction of foreign DNA into the host genome that changes the genetic makeup of the organism and modifies gene functions. The main interest of plant breeders is to manipulate genes for the development of biotic and abiotic stress-resistant varieties (Kumar et al., 2015). Recent advances in the development of transgenic cumin may enhance commercial value and its quality in the future. Recombinant traits have been established using embryos of cumin through microprojectile (Pandey et al., 2013). Agrobacterium-mediated genetic transformation and regeneration of plants using embryos as explant are reported in cumin, and genetically engineered salt-tolerant genotypes are also reported (Pandey et al., 2016; Niazi et al., 2019).

It may be useful for the cultivation of cumin in saline-prone areas. A saline gene known as *SbNHX1*, cloned from an extreme halophyte, *Salicornia brachiata*, was transformed in cumin using *Agrobacterium*, and RT-PCR analysis confirmed the presence of the gene. In salinity stress conditions, transformed cumin shows lower electrolytic leakage, lipid peroxidation, and proline content and higher photosynthetic pigments as compared to wild-type plants.

Agrobacterium-mediated gene transformation technology also creates salinity tolerance in other spice crops like *Trachyspermum ammi*, using betaine aldehyde dehydrogenase (BADH) gene. A study based on

bioassay revealed enhanced salinity tolerance in *Trachyspermum ammi*. The gene transformation is applicable for other Apiaceae family members for different purposes (Bhuyan et al., 2023).

3.1.4. Genome editing

In the last few decades, regular uses of transgenic tools have had their own limitations due to taking a longer time to transfer desirable traits. In addition, gene transfer via conventional methods integrates DNA at any random sites in the genome, leading to the suppression of functional genes, which results in a change in the plant's agronomic traits. Based on recent studies, genome editing emerges as a new strategy for improvement in spices also. Gene editing tools insert DNA into specific sites of the genome. In addition, in recent years, new genome editing-based approaches, such as targeted genome editing, can generate custom-designed medicinal plants with different secondary metabolite profiles. In recent years, efficient gene editing tools have been developed.

Meganucleases, Transcription activator-like effector nucleases (TALENs), Zinc finger nucleases (ZFNs), and Cas proteins are the different genome editing technologies used (Hilioti et al., 2016). ZFNs are able to induce homologous recombination and target DNA insertion in plants. In one study, it is found that customized ZFNs successfully cleaved the target site and induced mutations in the *LIL4* gene of tomato and expressed in tomato seeds that were confirmed by phenotypic variation in tomatoes (Shukla et al., 2009). In another study, ZFNs were used to insert a herbicide tolerance gene into inositol-1,3,4,5,6-pentaphosphate 2-kinase (IPK1), making it inactive (Alamillo et al.,

2023). Even though the use of these two technologies made an immense impact on crop improvement, certain limitations pave the way for (clustered regularly interspaced short palindromic repeats-associated) CRISPR/Cas system (Singh et al., 2023). The genome of an organism can be manipulated in a specific way with CRISPR/Cas system. The CRISPR/Cas system consists of guide RNA and Cas nuclease, which create DNA double-stranded breaks at specific nucleotide sequences on the target genome (Collonnier et al., 2017). Genome modification can be obtained by homologous recombination or end joining of foreign DNA, resulting in targeted transgene insertion or replacement (Angon et al., 2023). The recognition site specificity can be changed by altering the variable region of the guide RNA, making CRISPR/Cas a highly programmable tool.

CRISPR/Cas is a revolutionary tool for editing genes at specific locations of the genome as well as regularly used for biosystematics studies in crop plants. This process not only alters complex biotic & abiotic traits during crop improvement programs but also develops climate-resilient genotypes (Karavolias et al., 2021; Meena et al., 2021). Moreover, technology has various applications in crop improvement programs, genetic manipulation, deleting undesirable traits or introducing desired traits, and the least contamination of natural ecosystems. Thus, CRISPR/Cas is a potential tool for worldwide crop production and environmental sustainability. (Zhou et al., 2021). Currently, various applications of technology in medicinal plants mainly focus on model plants with complete genome information and efficient genetic transformation systems. knockout of laccase genes through CRISPR/Cas9 in *Salvia miltiorrhiza* and *D. officinale* revealed its roles in growth, development, and metabolism (Kui et al., 2017). Now, technology is being regularly used for crop improvement of *D. officinale* (Gao et al., 2020). CRISPR technology is used to investigate cannabinoid synthesis and genetic improvement in *Cannabis* (Aparna, 2023). In that view, the present review proposed the various applications of these novel techniques for crop improvement of many spice crops.

4. Major thrust areas and future prospective

Most of the cumin genotypes released so far have been developed through conventional approaches. The development of a new cultivar of cumin using Marker-Assisted Breeding (MAS) is required to be explored further. Moreover, accurate characterization of functional traits in cumin needs to be drawn to our attention for the speed breeding program. In this regard, there is a need to develop an adequate research base for in-depth characterization of the genetic base at the biochemical and molecular level. In addition, employing new techniques like micropropagation, somaclonal variation, genetic transformation, gene editing, and molecular characterization of economically important accessions of cumin. However, the development of transgenic genotypes is further delayed due to the tedious regulatory approval from the government and the high upfront cost. Moreover, the implications of advanced breeding tools required sophisticated infrastructure and complex regulatory policies. Advanced breeding techniques can significantly speed up the process of breeding and tackle issues like development of disease resistance and climate-resilient cultivars (Table 4). Currently, CRISPR/Cas9 has become a promising nuclease system and is widely used to produce biotic and abiotic stress-tolerant plants (Angon et al., 2023) because it is easy to set up and to design targets and constructs. Further, the use of these techniques will bridge the technological gap soon. In addition, these modern biotechnological methods are to be used for enhancing the therapeutic properties of cumin. Technology may increase the availability of various known bioactive phytochemical compounds like antioxidant agents, sesquiterpenoids, and flavonoids, which have potential anti-diabetic and anticancerous activities as mentioned in Table 3 (Nouri et al., 2024; Mohammed et al., 2024). In the future, genome editing will be the valuable tool for the improvement of chemical composition, biological activities, stress tolerance properties, and yield quality traits of cumin across the globe (Fig. 2) and the use of

genome editing tools may further act as a game changer for commercial cultivation. Properties of bioactive phytochemical compounds in cultivated species of cumin may further upgrade using conventional as well as biotechnological tools. Currently, modern tools not only enhance the global production of cumin but also value chain addition for cumin production. Integrated approaches not only balance the nutrient requirements but also play a major role in enhancing the secondary metabolite production. Moreover, congruent use of modern tools may further enhance essential oil and phytochemicals compounds, which ultimately affect the quality production of cumin. Therefore, in future characterization of potential landrace genotypes and introduction of ideotype breeding, may enhance the production of cumin as per global demand.

5. Conclusion

Cumin is routinely used in the kitchen as well as in Ayurvedic medicine. It consists of essential oil, which is mainly used in the food industry as well as for cosmetic, perfume, and pharmaceutical applications. Currently, the popularity of cumin, along with other spices and condiments, has risen, which has led to increased demand in the global market. India is a global leader in the cultivation and consumption of cumin. The present review suggested the integrated use of conventional and modern breeding tools for the improvement of cumin crops. This study concluded multidisciplinary approaches with novel gene editing tools provide the opportunity to transfer desirable traits in cumin genotypes for improvement. In the future, the development of a crop-specific sequencing database of cumin may further be useful to characterize stress-tolerant traits at the molecular level. The study discussed the integrated use of genomics, proteomics and biotechnological approaches to provide a better platform for the identification and characterization of genes encoding biologically active compounds and secondary metabolite production in cumin.

CRedit authorship contribution statement

Sharma Kirti: Writing – original draft, Data curation, Conceptualization. **Kumar Sanjeev:** Writing – review & editing, Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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